

06_Class_Activity

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In class activity 6:

What did we do last time in activity 5?

- Understanding standard normal distributions and z-scores
- Calculating and interpreting standard error
- Creating confidence intervals
- Working with the Student's t-distribution

Today's focus:

- Review more r code
- understand α alpha and β beta errors
- do more
 - 1 sample t tests
 - 2 sample t tests

Goes with Lecture 6

```
# Install packages if needed (uncomment if necessary)
# install.packages("readr")
# install.packages("tidyverse")
# install.packages("car")
# install.packages("here")

# Load libraries
library(patchwork)
library(car)           # For diagnostic tests
```

Loading required package: carData

```
library(tidyverse)    # For data manipulation and visualization
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4      ✓ readr      2.1.5
✓ forcats    1.0.0      ✓ stringr    1.5.1
✓ ggplot2    3.5.2      ✓ tibble     3.3.0
✓ lubridate  1.9.4      ✓ tidyr      1.3.1
✓ purrr      1.1.0
```

```
— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
✖ dplyr::recode()  masks car::recode()
```

```
* purrr::some() masks car::some()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Load the pine needle data
# Use here() function to specify the path
pine_data <- read_csv("data/pine_needles.csv")
```

```
Rows: 48 Columns: 6
— Column specification —————
Delimiter: ","
chr (4): date, group, n_s, wind
dbl (2): tree_no, length_mm

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Examine the first few rows
head(pine_data)
```

```
# A tibble: 6 × 6
  date      group      n_s  wind tree_no length_mm
<chr>   <chr>    <chr> <chr>  <dbl>    <dbl>
1 3/20/25 cephalopods n    lee      1        20
2 3/20/25 cephalopods n    lee      1        21
3 3/20/25 cephalopods n    lee      1        23
4 3/20/25 cephalopods n    lee      1        25
5 3/20/25 cephalopods n    lee      1        21
6 3/20/25 cephalopods n    lee      1        16
```

Part 1: Exploratory Data Analysis

Before conducting hypothesis tests, we should always explore our data to understand its characteristics.

Let's calculate summary statistics and create visualizations.

Activity: Calculate basic summary statistics for pine needle length

```
# YOUR TASK: Calculate summary statistics for pine needle length
# Hint: Use summarize() function to calculate mean, sd, n, etc.

# Create a summary table for all pine needles
pine_summary <- pine_data %>%
  group_by(wind) %>%
  summarize(
    mean_length = mean(length_mm, na.rm=TRUE),
    sd_length = sd(length_mm, na.rm=TRUE),
    n = sum(!is.na(length_mm)),
    se_length = sd_length / (n^0.5),
    t_critical = qt(0.975, df = n - 1), # 95% CI uses 0.975 (two-tailed)
    ci_lower = mean_length - t_critical * se_length,
    ci_upper = mean_length + t_critical * se_length
  )
```

```
print(pine_summary)
```

```
# A tibble: 2 × 8
  wind mean_length sd_length    n se_length t_critical ci_lower ci_upper
<chr>    <dbl>    <dbl> <int>    <dbl>    <dbl>    <dbl>    <dbl>
1 lee      20.4      2.45    24    0.500      2.07     19.4     21.5
2 wind     14.9      1.91    24    0.390      2.07     14.1     15.7
```

```
# Now calculate summary statistics by wind exposure
# YOUR CODE HERE
```

Part 1: Visualizing the Data

Activity: Create visualizations of pine needle length

Create a histogram and a boxplot to visualize the distribution of pine needle length values.

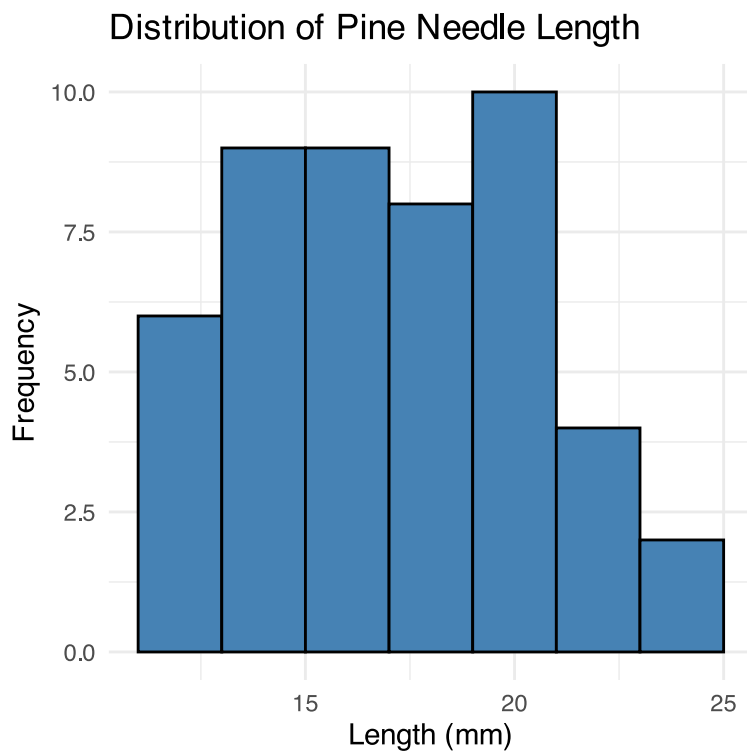
Effective data visualization helps us understand:

- The central tendency
- The spread of the data
- Potential outliers
- Shape of distribution

Your Task

```
# YOUR TASK: Create a histogram of pine needle length
# Hint: Use ggplot() and geom_histogram()

# Histogram of all pine needle lengths
ggplot(pine_data, aes(x = length_mm)) +
  geom_histogram(binwidth = 2, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Pine Needle Length",
       x = "Length (mm)",
       y = "Frequency") +
  theme_minimal()
```



```
# how can you do this by wind to see both plots
```

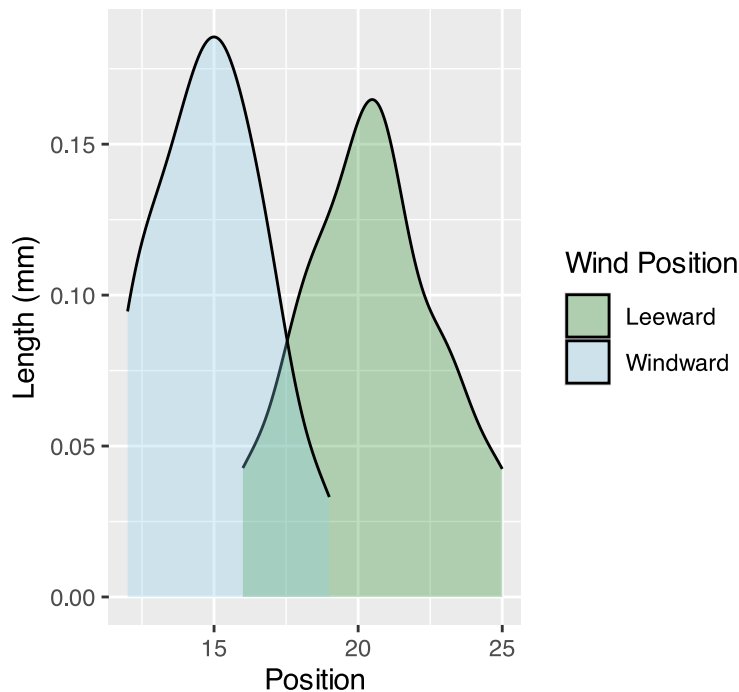
```
# Boxplot of pine needle length by wind exposure
# YOUR CODE HERE
```

Can you plot the density distributions for the two samples

```
# Histogram of all pine needle lengths
pine_data %>%
  ggplot(aes(x = length_mm, fill=wind)) +
  geom_density(color = "black", alpha=0.3, trim = TRUE) +
  labs(title = "Pine Needle Lengths by Wind Exposure",
       x = "Position",
       y = "Length (mm)",
       fill = "Wind Position") +

  scale_fill_manual(values = c("lee" = "forestgreen", "wind" = "skyblue"),
                    labels = c("lee" = "Leeward", "wind" = "Windward"))
```

Pine Needle Lengths by Wind Exposure



what is the Effect size or difference in means?

💡 Practice Exercise: Calculate Effect size

We could also look at the difference in means... some cool code here

```
# Assuming your dataframe is called df
pine_summary %>%
  summarize(difference = mean_length[wind == "lee"] - mean_length[wind == "wind"])
```

```
# A tibble: 1 × 1
  difference
  <dbl>
1       5.5
```

Part 1: Two Sample T-Test

Now, let's compare pine needle lengths between windward and leeward sides of trees.

Question: **Is there a significant difference in needle length between the windward and leeward sides?**

This requires a two-sample t-test.

Two-sample t-test compares means from two independent groups.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where:

- $\bar{x}_1$ and $\bar{x}_2$: These represent the sample means of the two groups you're comparing
- s_p^2 : This is the pooled variance, calculated as: $s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$, where s_1^2 and s_2^2 are the sample variances of the two groups.

- n_1 and n_2 : These are the sample sizes of the two groups.
- $\sqrt{(1/n_1 + 1/n_2)}$: This represents the pooled standard error.

Part 1: Testing Assumptions for Two-Sample T-Test

Activity: Test assumptions for two-sample t-test

For a two-sample t-test, we need to check:

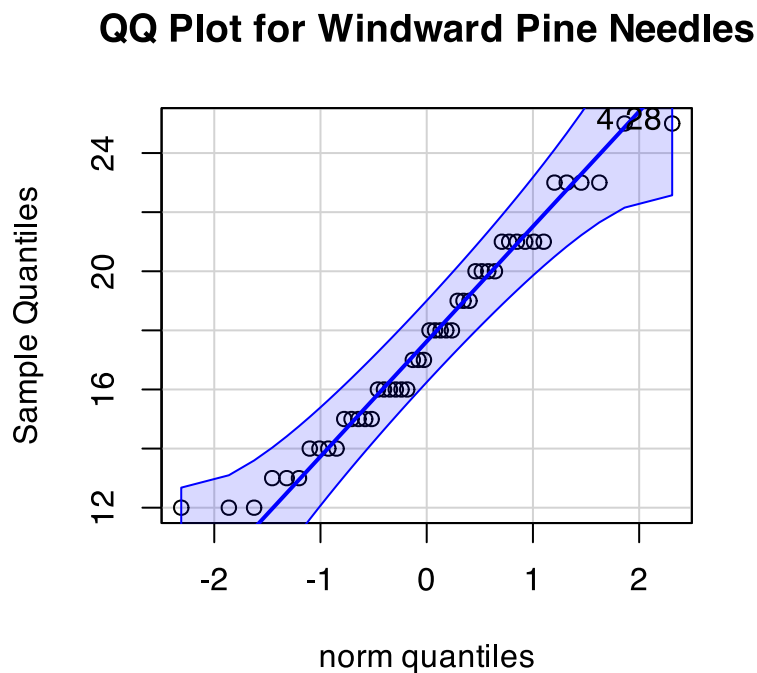
1. Normality within each group
2. Equal variances between groups (for standard t-test)
3. Independent observations

If assumptions are violated:

- Welch's t-test (unequal variances)
- Non-parametric alternatives (Mann-Whitney U test)

Your task

```
# YOUR TASK: Test normality of windward pine needle lengths
# QQ Plot
qqPlot(pine_data$length_mm,
       main = "QQ Plot for Windward Pine Needles",
       ylab = "Sample Quantiles")
```



```
[1] 4 28
```

```
# Testing normality for each group
# Leeward group
lee_data <- pine_data %>% filter(wind == "lee")
```

```
shapiro_lee <- shapiro.test(lee_data$length_mm)
print("Shapiro-Wilk test for leeward data:")
```

```
[1] "Shapiro-Wilk test for leeward data:"
```

```
print(shapiro_lee)
```

Shapiro-Wilk normality test

```
data: lee_data$length_mm
W = 0.95477, p-value = 0.3425
```

windward group

```
# Windward group
# YOUR CODE HERE for windward group normality test
```

Remember you can always do it in one go

```
# there are always two ways
# Test for normality using Shapiro-Wilk test for each wind group
# All in one pipeline using tidyverse approach
normality_results <- pine_data %>%
  group_by(wind) %>%
  summarize(
    shapiro_stat = shapiro.test(length_mm)$statistic,
    shapiro_p_value = shapiro.test(length_mm)$p.value,
    normal_distribution = if_else(shapiro_p_value > 0.05, "Normal", "Non-normal")
  )

# Print the results
print(normality_results)
```

```
# A tibble: 2 × 4
  wind shapiro_stat shapiro_p_value normal_distribution
<chr>      <dbl>          <dbl> <chr>
1 lee      0.955          0.343 Normal
2 wind     0.961          0.451 Normal
```

Conduct a Levene's Test

```
# Test for equal variances
# YOUR TASK: Conduct Levene's test for equality of variances
levene_test <- leveneTest(length_mm ~ wind, data = pine_data)
```

```
Warning in leveneTest.default(y = y, group = group, ...): group coerced to
factor.
```

```
print(levene_test)
```

```
Levene's Test for Homogeneity of Variance (center = median)
  Df F value Pr(>F)
group 1  1.2004 0.2789
    46
```

```
# Visual check for normality with QQ plots
# YOUR CODE HERE
```

Part 2: Conducting the Two-Sample T-Test

Activity: Conduct a two-sample t-test

Now we can compare the mean pine needle lengths between windward and leeward sides.

$H_0: \mu_1 = \mu_2$ (The mean needle lengths are equal)

$H_1: \mu_1 \neq \mu_2$ (The mean needle lengths are different)

Deciding between:

- Standard t-test (equal variances)
- Welch's t-test (unequal variances)

Based on our Levene's test result.

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

# Standard t-test (if variances are equal)
t_test_result <- t.test(length_mm ~ wind, data = pine_data, var.equal = TRUE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```
print(t_test_result)
```

Two Sample t-test

```
data: length_mm by wind
t = 8.6792, df = 46, p-value = 3.01e-11
alternative hypothesis: true difference in means between group lee and group wind is not equal to 0
95 percent confidence interval:
 4.224437 6.775563
sample estimates:
mean in group lee mean in group wind
    20.41667      14.91667
```

```
# Calculate t-statistic manually (optional - uggg - maybe )
# YOUR CODE HERE: t = (mean1 - mean2) / sqrt((s1^2/n1) + (s2^2/n2))
```

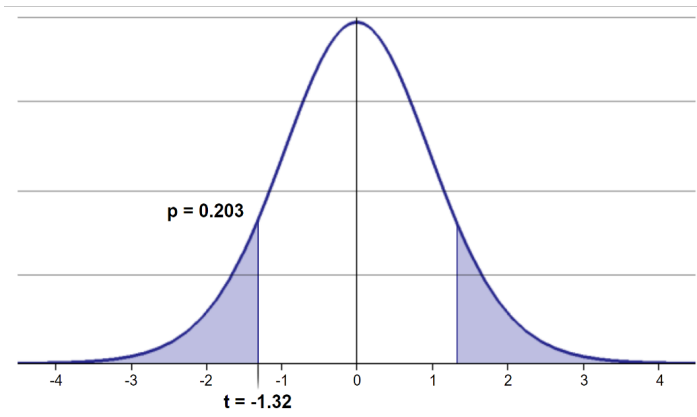

Part 2: Interpreting and Reporting Two-Sample T-Test Results

Activity: Interpret the results of the two-sample t-test

What can we conclude about the needle lengths on windward vs. leeward sides?

How to report this result in a scientific paper:

“A two-tailed, two-sample t-test at $\alpha=0.05$ showed [a significant/no significant] difference in needle length between windward ($M = \dots$, $SD = \dots$) and leeward ($M = \dots$, $SD = \dots$) sides of pine trees, $t(\dots) = \dots$, $p = \dots$ ”



What is Power

Statistical power represents the probability of detecting a true effect (rejecting the null hypothesis when it is false). In this case, with a power of 97%, there's a 97% chance of detecting a true difference of 30 units between the means of the two groups if such a difference actually exists.

A power analysis like this is typically done for one of these purposes:

1. Before data collection to determine required sample size
2. After a study to evaluate if the sample size was adequate
3. To determine the minimum detectable effect size with the given sample

With 97% power, this test has excellent ability to detect the specified effect size. Generally, **80% power is considered acceptable**, so 97% indicates a very well-powered study for detecting a difference of 30mm between the groups.

```
lee_df <- pine_data %>% filter(wind == "lee")
wind_df <- pine_data %>% filter(wind == "wind")
# Calculate power for detecting a 1 mm difference
wind_diff = 1

lee_n <- nrow(lee_df)
wind_n <- nrow(wind_df)

wind_sd_pooled <- sqrt((var(lee_df$length_mm) * (lee_n-1) +
                        var(wind_df$length_mm) * (wind_n-1)) /
                      (lee_n + wind_n - 2))

# Calculate power
wind_effect_size <- wind_diff / wind_sd_pooled # Cohen's d
wind_df <- lee_n + wind_n - 2
wind_alpha <- 0.05
wind_power <- power.t.test(n = min(lee_n, wind_n),
                          delta = wind_effect_size,
                          sd = 1, # Using standardized effect size
                          sig.level = 0.5,
```

```

type = "two.sample",
alternative = "two.sided")

# Display results
wind_power

```

Two-sample t test power calculation

```

n = 24
delta = 0.4555423
sd = 1
sig.level = 0.5
power = 0.8158402
alternative = two.sided

```

NOTE: n is number in *each* group

Now to make a final plot

Typically we will make a plot that has the mean and standard error on it to represent the data

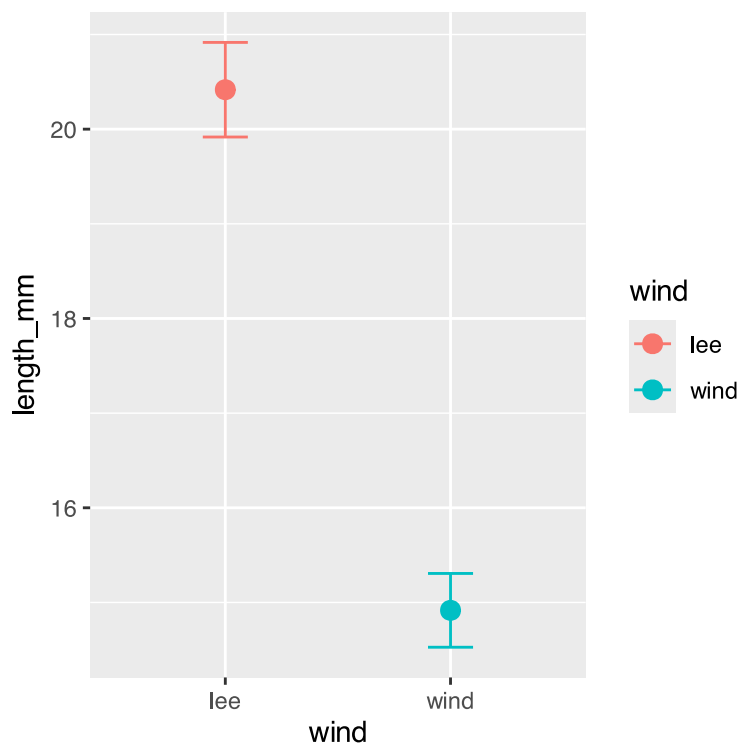
your Task is to make this plot

```

pine_mean_se <- pine_data %>%
  ggplot(aes(wind, length_mm, color = wind))+
  stat_summary(fun = "mean", na.rm=TRUE, geom="point", size = 3)+
  stat_summary(fun.data = "mean_se", width = 0.2, geom = "errorbar")

pine_mean_se

```



Summary and Conclusions

In this activity, we've:

1. Formulated hypotheses about pine needle length
2. Tested assumptions for parametric tests
3. Conducted a two-sample t-tests
4. Visualized data using appropriate methods
5. Learned how to interpret and report t-test results

Key takeaways:

- Always check assumptions before conducting tests
- Visualize your data to understand patterns
- Report results comprehensively
- Consider alternatives when assumptions are violated

Reflection Questions

After completing the activities, discuss these questions with your group:

1. How does sample size affect our confidence in estimating the population mean?
2. Why is the t-distribution more appropriate than the normal distribution when working with small samples?
3. When comparing two populations, what can we learn from looking at confidence intervals versus performing a t-test?
4. How would you explain the concept of statistical significance to someone who has never taken a statistics course?
5. What do we do if assumptions FAIL!!!